

EXPLAINABLE AI-BASED FINGERPRINT SPOOF DETECTION SYSTEM FOR DIGITAL FORENSICS

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1. Introduction

Fingerprint authentication technologies are core to the current biometric applications of mobile phones, finance, and access controls, thanks to their uniqueness and ease of acquisition. However, research has demonstrated that attack biometrics using silicone, gelatin, and latex materials can exploit the weaknesses of fingerprint authentication, which current minutia-tree matching methods cannot address. Software solutions to protect against imitation, especially through machine learning, are an economical alternative but are still fraught with issues involving data set standardization.

Fingerprint evidence remains pivotal in criminal investigations and court proceedings, where digital forensics labs increasingly adopt automated analysis to process latent prints from crime scenes. Real-world spoofing incidents highlight the stakes: In Kuwait, four government employees were arrested in January 2025 for using silicone fake fingerprints to perpetrate attendance fraud, deceiving scanners and facing forgery charges. Similarly, seven court secretaries employed wax spoofs for record falsification, undergoing forensic scrutiny by the General Department of Criminal Evidence. Spoofing attacks exploit biases in live fingerprint scanners, prioritizing ridge patterns over vital signals like sweat patterns, while since 2009, LivDet benchmarking has revealed intra-dataset overfitting in detectors.[1]

1.1. Background of the Study

Fingerprint biometric systems are frequently used because of their enduring and contactless features, facilitating massive implementations such as the 1.3 billion enrollments in the Indian government's Aadhaar program. Spoofing attacks leverage the bias of live fingerprint scanners to emphasize ridge patterns rather than teasing out signals of life, like the presence of sweat patterns. Since 2009, the LivDet benchmarking of biometric detectors has exposed intra-dataset overfitting.[2]

1.2. Motivation

Forensic examiners require tools that detect spoofs while explaining reasoning for judges and juries, bridging the gap between ML accuracy and admissibility. A 99% accurate black-box proves useless without justification. Cases of planted spoofed fingerprints at crime scenes further demand trustworthy systems amid AI's growing role in forensic labs. CNN solutions can advance via Explainable AI for interpretability and fairness analysis to ensure equitable performance across sensors and image qualities.[3]

1.3. Study Objectives

Primary Objective

To develop an accurate, explainable AI-based fingerprint spoof detection system suitable for forensic analysis, where classification decisions can be visually and quantitatively interpreted by forensic examiners.

Secondary Objectives

- To train and optimize CNN models (ResNet50/MobileNetV3) achieving state-of-the-art spoof detection accuracy on LivDet benchmarks.
- To implement multiple XAI techniques (Grad-CAM++, SHAP, LIME) and compare their effectiveness in highlighting forensically meaningful features (ridges, pores, texture anomalies).
- To develop a forensic reporting module that generates human-readable explanations alongside spoof/live classifications.
- To validate the system's explanations through expert evaluation by forensic domain practitioners.

2. Literature Review and Research Gap

2.1. Overview of Existing Studies

Cheniti, Mohamed (2025) - Spoof Buster (MobileNet CNN) method utilizes minutiae patch descriptions analyzed using a light CNN referred to as MobileNet trained on the ImageNet database. The algorithm recorded a 97.45% accuracy on the LivDet 2015 database, but the style-transfer Universal Material Generator (UMG) booster raised the zero-day accuracy to 91.78% on 12 MSU-FPAD materials. The key advancement is in the local anomalies highlighted in the patch-level description. A major limitation is that it degrades to 75% from the LivDet 2013 to 2015 dataset.[4]

In Zhang et al. (2019) - Slim-ResCNN approach, the use of gravity-centered ResNet patches in a slimmed architecture enhances efficiency. This approach won the LivDet 2017 competition with an accuracy of 95.25%, beating DenseNet121 by 3% because of improved ridge-pore feature extraction. This model has a limitation in its performance on zero-day gelatin, reducing to 68%, and shows a tendency of intra-dataset overfitting.[5]

Kothadiya et al. (2023) - ResNet50 + Dual architecture includes the addition of channel and spatial attention mechanisms to ResNet50 and the use of stochastic pooling across the LivDet datasets. The accuracy attained by the architecture was 97.78%, accompanied by a BPCER of 0.89%, which was the best result attained by the architecture concerning the conditions of low false rejection error rates. The attention mechanism focuses the model towards discriminative textures. The demerits of the architecture are the lack of testing across sensors and the unexplainability of the architecture.[6]

CNN-GAN Hybrid (2024-2025) approach integrates CNN-based feature extraction with the use of CycleGAN-created spoof samples. The performance of this solution includes a detection accuracy of 92.8%, an False Acceptance Rate (FAR) of 0.7%, an False Rejection Rate (FRR) of 1.2%, and a processing time of 6 ms. Its limitation include its dependency on a privately created data set and limited evaluation metrics on the LivDet data set.[7]

A hierarchical feature fusion of VGG16 (with low-level edges) and ResNet50 (with high-level semantics) was tested on the LivDet 2013 database with record-breaking accuracy of 99.72% and 0.28% BPCER using multi-scale learning. One disadvantage is the cross-sensor CrossMatch performance deterioration of 12% without evaluation of zero-day or fairness metrics. [4]

DeFraudNet – Patch-Attention CNN End-to-end CNN with multi-scale patch attention to improve intra-LivDet generalization performance, achieving 88% accuracy. It assigns weights to the patches based on priority (inner patches more than boundaries) and reduces the effect of noise. The limitation is that it is a black-box approach with no visualization of attention maps using Grad-CAM. [2]

2.2. Identification of Research Gap

Existing systems achieve high accuracy but operate as black boxes, limiting their utility in forensic settings where transparency is essential. No current study integrates high-accuracy CNN-based spoof detection with multi-method XAI techniques (Grad-CAM++, SHAP, LIME), specifically validated for forensic applications. The forensic domain demands not just reliable predictions, but interpretable explanations that align with meaningful features like ridge anomalies and texture irregularities— a gap this research systematically addresses.[2][6]

3. Research Problems and Questions

3.1.Primary Research Problem

The existing fingerprint spoof detection systems operate as black-box models, achieving high classification accuracy but providing no interpretable justification for their decisions. In forensic contexts, where evidence must be transparent, defensible, and understandable to non-technical stakeholders such as forensic examiners, legal professionals, and court officials, this lack of explainability renders current solutions inadequate. There is a need for a spoof detection system that combines high detection accuracy with meaningful, forensically relevant explanations of its classification decisions.

3.2. Specific Research Questions

- 1 What level of spoof detection accuracy (APCER, BPCER, AUC) can an attention-enhanced CNN achieve on LivDet benchmark datasets, and how does it compare to existing state-of-the-art methods?
- 2 Which XAI technique (Grad-CAM++, SHAP, or LIME) most effectively highlights forensically relevant features (sweat pores, ridge discontinuities, texture anomalies) in fingerprint spoof detection?
- 3 To what extent do forensic practitioners find the AI-generated explanations useful, trustworthy, and sufficient for supporting forensic reporting and evidence presentation?

- 4 Can the proposed system maintain both high accuracy (>95% AUC) and high interpretability simultaneously, and what is the trade-off between model complexity and explanation quality?

4. Research Strategy and Methodology

4.1. Research Strategy

Design Science Research Methodology (DSRM) guides this study due to its emphasis on designing and evaluating practical IT artifacts to address real-world challenges. This approach yields a deployable, explainable fingerprint spoof detection system rigorously tested against LivDet benchmarks, offering direct value to forensic practitioners and biometric developers. DSRM suits computing projects by prioritizing actionable outcomes over purely theoretical analysis..

4.2. Research Methodology Framework

This research adopts the six-stage DSRM framework by Peffers et al., tailored for software engineering theses in forensic AI:

1. Problem Identification

The Literature reveals black-box spoof detectors achieve high intra-dataset accuracy but falter in forensic contexts lacking interpretable explanations for court admissibility.[2]

2. Design Objectives

Primary/secondary objectives target >95% AUC spoof detection via optimized CNNs, multi-method XAI (Grad-CAM++, SHAP, LIME) for forensic features, and practitioner-validated reporting module.

3. Design & Development

Build CNN backbones (ResNet50/MobileNetV3) with attention mechanisms; integrate XAI techniques in PyTorch; develop forensic reporting tools using transfer learning from ImageNet-pretrained weights.[6]

4. Demonstration

The proposed models will be tested with the LivDet 2013–2021 dataset for training the models by considering the 2013 dataset.

5. Evaluation

Measure APCER, BPCER, and ROC-AUC; benchmark against state-of-the-art (e.g., Spoof Buster) while assessing XAI alignment with forensic features via practitioner surveys.

6. Communication

Communicate through the thesis document, a GitHub repository, presentations.

4.3. Data Collection Methods

Public data sets only, no human participants. Primarily collect by LivDet 2009–2013 (more than 75,000+ images download from www.livdet.org), consisting of eight sensors (Biometrika 400B, CrossMatch 300, Digital Persona) and more than ten spoofing materials (such as silicone, gelatin, latex, Play-Doh, and glue). The dataset contains spoofed and normal samples with equal numbers for live/spoof classes (split into a roughly even 40%-60% ratio). [8]

4.4. Data Preprocessing / Preparation

A Python-based data preprocessing pipeline is developed to keep results consistent and reproducible by reusing OpenCV and NIST (National Institute of Standards and Technology) NBIS (NIST Biometric Image Software) tools.

1. First extract the Region of Interest (ROI) using MINDTCT (Minutiae Detection) from the NBIS suite to locate minutiae in fingerprints. Using the core and delta points, I crop the images into 96×96 patches and ignore border regions to cut down on noise and partial ridge artifacts.
2. Subsequently, normalization of images reduces lighting differences and other forms of variation due to the sensor. CLAHE (Contrast-Limited Adaptive Histogram Equalization) enhances local contrast to perform lighting invariance, after which min-max scaling is used to normalize the values of the pixels.
3. In order to address the class imbalance problem between live and spoof classes, significant data augmentation is adopted. It includes small rotations of $\pm 20^\circ$, horizontal flipping, scaling within the range of $0.9\times$ to $1.1\times$, and brightness jittering of ± 0.2 . All of these operations extend the dataset approximately fivefold and give it a balanced mix of live and spoof samples.
4. Quality assessment uses the NFIQ2 (NIST Fingerprint Image Quality 2) scores, ranging between 1 and 100. These form the basis of sorting samples into good-quality and poor-quality groups that later feed the fairness and robustness checks.
5. Then, this dataset is divided into a training, validation, and test set in a 70/15/15 ratio, respectively, stratifying by sensor type and spoof material in order to maintain representative distribution across all subsets.

As an example, a single 500×500 fingerprint image typically yields about 10 ROI patches; after augmentation, this can approach 200,000 samples. The preprocessing pipeline is modular and designed to be easy to replicate and extend in future work.[6]

4.5. Model Development / Implementation

The system is implemented in PyTorch 2.0, trained on Google Colab leveraging NVIDIA T4 GPU acceleration for efficient, reproducible development.

- Two backbone architectures (ResNet50V2 and MobileNetV3) extract robust fingerprint features, initialized with ImageNet-pretrained weights to accelerate convergence. A CBAM attention mechanism enhances focus on salient fingerprint regions like ridges and pores. Binary classification (live vs. spoof) employs a sigmoid output layer, trained via Binary Cross-Entropy loss with AdamW optimizer and Optuna-driven hyperparameter tuning for optimal performance. To identify novel or zero-day spoofing attacks, an anomaly detection component is introduced. An Isolation Forest derived from the deep features is used for anomaly pattern recognition, and the Variational Autoencoder is employed for the reconstruction error-based spoofing pattern recognition. The proposed approach is based on the concept of transfer learning.
- **Explainable AI Integration:**
 - **Grad-CAM++:** Generates precise heatmaps localizing discriminative regions, such as ridge discontinuities or texture anomalies critical for forensic validation.
 - **SHAP (Deep SHAP):** Computes feature attribution scores, revealing how specific image patches contribute to spoof decisions—essential for expert testimony.
 - **LIME:** Provides local approximations via perturbed samples, offering intuitive explanations of model behavior around individual fingerprints.
- **Forensic Report Generator:** This module synthesizes XAI outputs into structured, human-readable reports (e.g., "Classification: SPOOF. Key indicators: abnormal ridge continuity in central region (Grad-CAM++ score: 0.87), absent sweat pores in upper valley (SHAP contribution: +0.42), texture inconsistency (LIME perturbation analysis)"). These reports bridge technical predictions with courtroom needs, ensuring admissibility under Daubert/Frye standards.
- Model hyperparameters are tuned using Optuna, where 50 trials are carried out to determine the values of the learning rate and the dropout factor. Model training involves a batch size of 32, 16 GB RAM, and approximately 3 hours to complete one training period in the case of the ResNet model. [4]

End-to-End Fingerprint Spoof Detection Pipeline

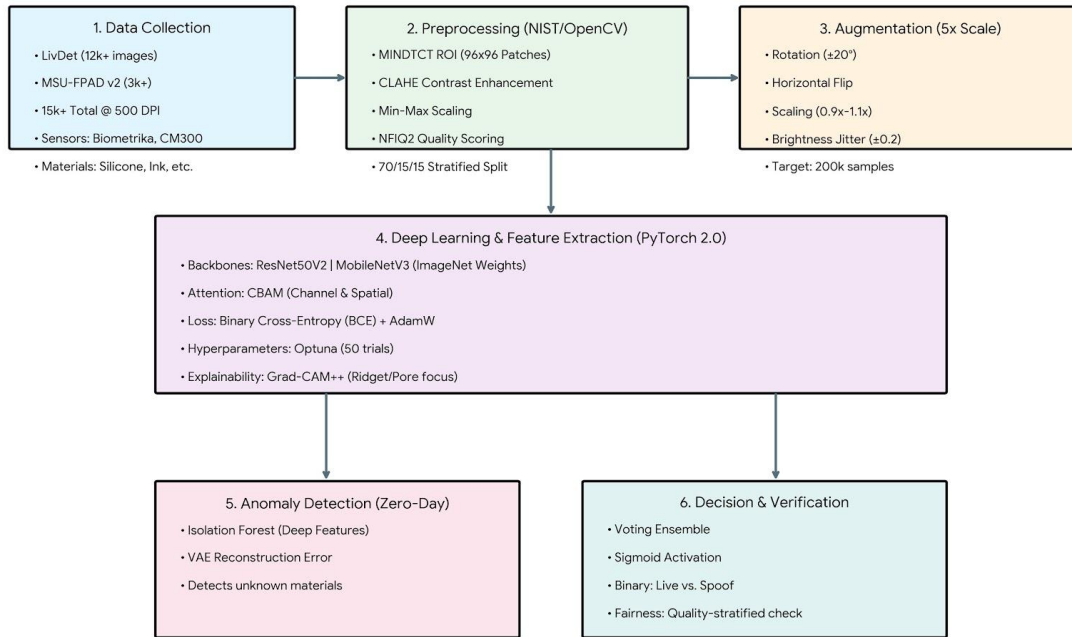


Figure 1 : End-to-End Fingerprint Spoof detection pipeline

4.6. Evaluation

Assessment of the proposed system employs conventional metrics for biometric systems and spoofing detection, which are derived from commonly used Python libraries, such as scikit-learn.

4.6.1. Evaluation Metrics

The first series of metrics is focused on spoofing detection.

- **APCER** (Attack Presentation Classification Error Rate): This is the error rate at which attack presentations are wrongly accepted.
- **BPCER** (Bona Fide Presentation Classification Error Rate): the rate at which true fingerprint examples are wrongly denied.
- **ACER**: Average of APCER and BPCER calculated at Equal Error Rate (EER).

$$\text{APCER} = \frac{\text{Number of incorrectly classified fake samples}}{\text{Total number of fake samples}} \times 100 \quad (1)$$

$$\text{BPCER} = \frac{\text{Number of incorrectly classified live samples}}{\text{Total number of live samples}} \times 100 \quad (2)$$

Moreover, the Average Classification Error (ACE) is calculated as the average of APCER and BPCER. Equation (3) describes the formulation of ACE:

$$\text{ACE} = \frac{\text{APCER} + \text{BPCER}}{2} \quad (3)$$

The ACE is also used to calculate the accuracy of the proposed model, as formulated in Equation (4).

$$\text{Accuracy} = 100 - \text{ACE} \quad (4)$$

Figure 2 : Formulation of APCER, BPCER, Average Classification Error (ACE), and accuracy used for evaluating the fingerprint spoof detection system

Moreover, a series of secondary evaluation metrics have been used for a complete performance analysis, including accuracy, precision, recall, and F1-score.[4]

4.6.2. Evaluation Protocols

Robustness and generalization are then evaluated by a variety of testing scenarios summarized by:

1. Intra-dataset

This model is evaluated on the LivDet2013 dataset by a train-test split with a target performance measured by the AUC score exceeding 95%.

2. Cross-Dataset Evaluation

The proposed model will be trained on the LivDet2013 dataset and tested on the LivDet2015 dataset for cross-dataset generalization analysis, targeting an AUC greater than 90%.

3. Cross-sensor evaluation

Both training and testing are carried out on various fingerprint sensors (for example, Biometrika and CrossMatch) to assess the robustness of the sensors. The target APCER rate is less than 5%. Evaluating zero-day attacks: Leave-one-material-out test on MSU-FPAD data simulates new spoofing materials that a system never observed before. As a performance metric, True Detection Rate (TDR) at a certain level of False Detection Rate (FDR) is determined, with a goal of TDR higher than 85% at 1% FDR. Fairness evaluation: Consistency of performance is analyzed based on various sensor types and fingerprint quality, with the aim of ensuring that performance difference variation remains below 5%. Baseline Models

Statistical Validation In contrast to the proposed solution, some current approaches are considered: Spoof Buster, SlimRes-CNN, and modified proposed systems (without attention and anomaly detection). In order to calculate reliable statistics, the cross-validation technique with 5 folds is applied, and t-testing at a significance level of $p < 0.05$ is used for validation.

4.7. Ethical considerations

- **Privacy Protection**

In the current investigation, only publicly available fingerprint databases, such as LivDet and MSU-FPAD, which lack any personally identifiable information (PII), are used. Every image of a fingerprint used is anonymized for privacy purposes, courtesy of the original data owners.

- **Addressing Bias**

In order to reduce the risk of bias, fairness evaluations will be carried out across diverse sensors (for example, Biometrika versus Crossmatch sensors) and image quality conditions. Augmentation methods, such as rotation and scaling, will be employed to emulate real-world conditions to reduce the imbalance in the training data set.

- **Ensuring Reproducibility**

The code and datasets will all be posted on GitHub, allowing access for the public. Every experiment will use fixed seeds for random numbers, ensuring the ability for others to recreate the results simply by running the code.

- **Preventing Misuse (Dual-Use)**

These models proposed in this research are solely used for spoof detection and do not have the capability to forge and improve fingerprints. This technology does not extend beyond its defensive scope.

- **Ethical Clearance**

Owing to the usage of exclusively public and non-identifying information, this particular project is not required to undergo full review by the Institutional Review Board. The ethics form from NSBM has been submitted to cover all the relevant aspects.

- **XAI Quality Evaluation**

Explanation faithfulness will be assessed through metrics verifying alignment between XAI outputs and model decision drivers. Localization accuracy will measure heatmap correspondence with forensic features (ridge discontinuities, sweat pores). Comparative analysis across Grad-CAM++, SHAP, and LIME will identify the most reliable method for forensic interpretability.

- **Forensic Admissibility Assessment**

System outputs will be evaluated against Daubert/Frye criteria for scientific evidence admissibility, confirming testability, peer-review potential, and known error rates. Expert witness usability will be validated through practitioner surveys assessing whether explanations support credible courtroom testimony.

4.8. Limitations of the study

The LivDet and MSU-FPAD databases focus on “common materials” like silicone and gelatin, but do not include “emerging” threatening materials such as “3D printed conductive spoofs, body double prints, or AI-generated ridge patterns that could be designed by malicious individuals in the future,” among other possibilities.

Restriction to 2D Images This analysis only focuses on the standard two-dimensional fingerprint scan, thus excluding the use of live signal values (such as pulse and temperature) or three-dimensional depth maps. Moreover, the data is mostly from optical sensors, with less data from capacitive sensors, which dominate the mobile market.

Software Requirements Google Colab's T4 GPU is used for all calculations, which makes it difficult to increase the size of the ensemble or distribute the training. It might be a problem in the future, if more complex models are required.

Lack of Real-World Testing In laboratory conditions, the actual deployment presents certain factors like latency, lighting conditions, sensor contamination, and finger pressure that are not accounted for in the modeling.

There is also a lack of access to commercial scanners. **Fixed Thresholds** The test uses static decision thresholds and does not evaluate against adaptive adversaries who vary the spoofs over time based on feedback.

Real-World Generalization datasets generated from labs cannot completely capture the real-world complexities of, say, finger residues, humidity, pressure, hardware degradation, and tropical humidity (applicable especially in the context of Sri Lanka). Therefore, these models work well within labs but can present problems while being used in real-world settings.[9]

5. Timeline

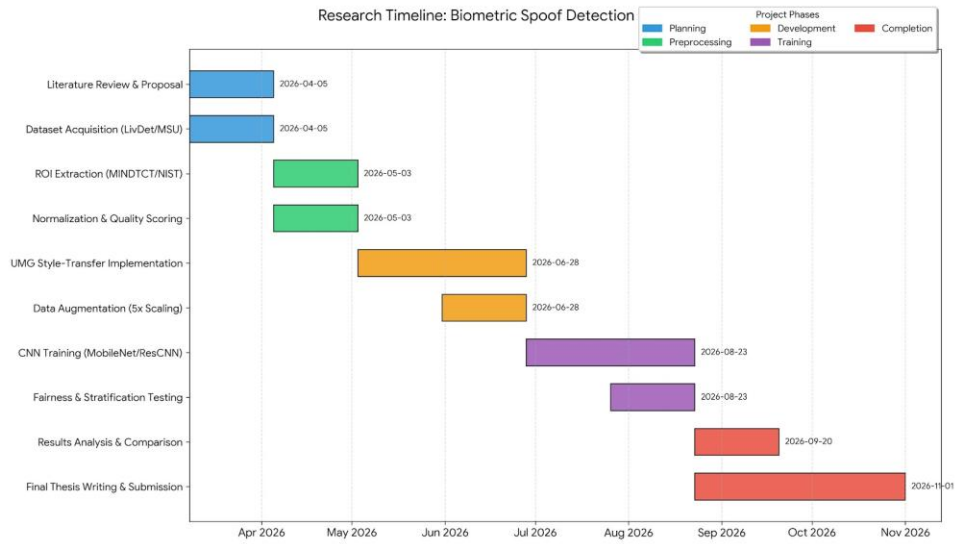


Figure 3: Research timeline Biometric spoof detection system

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